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Drawing approvals routed in order and chased, with invoice certification as the stretch

THE STORY

Document control at Ashvaka Infrastructure Limited runs one project mailbox for three live sites in Gujarat, where five contractors send in drawings for approval. Every drawing has to pass a fixed chain of reviewers in order, such as a design lead, a package manager and the client's consultant, and each has only a few days to act. Reviewers go on leave without telling the workflow, a step sits overdue for days, and a contractor sends revision C while revision B is still with the reviewers. Approvals and comments come back by email and as marked-up PDFs instead of in the system, and some emails do not even say which revision they are about. The AI Employee reads each transmittal, sends it to the right person at the right step, chases or redirects a stalled step, records every approval and comment against its revision, and asks the Project Director to name a stand-in when nobody is recorded.

OPEN THESE THREE FILES FIRST

- dataset/reference/Document_Control_Procedure.md** The rules; for Round 1 read sections 3 (routing and SLA) and 4 (leave and delegation) first.
- dataset/csv/Document_Register.csv** The work queue: one row per drawing, and its last_transmittal is the item in hand.
- dataset/csv/Workflow_Steps.csv** Shows which step each drawing is at, who holds it, when it is due and what was done.

WORDS YOU WILL MEET

Drawing A technical sheet (plan, section or detail) that the contractor must get approved before building from it. Each one has a number such as AIL-SND-CIV-DWG-0103.

Revision One version of a drawing. Letters (A, B, C) are used while it is in review and numbers (0, 1) once it is issued for construction, so Rev 0 is later than Rev C.

Transmittal The cover note a contractor sends with a drawing revision, with its own number such as TIL-AIL-TR-0202. Each new transmittal is a new item for the approval queue.

Document control The team that registers every drawing, routes it to reviewers, logs their comments and keeps track of which revision is the latest. Your AI Employee does this team's routine work.

Approval chain The fixed list of approvers, in order, for a work order and discipline. Step 2 is sent only after step 1 has acted.

SLA Service level agreement: the number of calendar days an approver has to act, counted from the day the step was sent. A step is overdue from the day after its due date.

Delegate A colleague an approver names in the leave calendar to act for them while away. Only a delegate recorded in the calendar counts, not one mentioned in an out-of-office reply.

PMC Project Management Consultant: the outside firm that represents the client and is usually the last step of the chain. It appears as Client Representative (PMC).

Approved with comments The approver accepts the revision but asks for changes. It counts as fully approved only once every comment on it is closed.

Marked-up file A PDF of the drawing that a reviewer has written comments on and sent back. Those comments count as much as comments typed in the workflow tool or sent by email.

Superseded A revision is superseded when a later revision of the same drawing has been issued. It does not mean the later one is approved yet, and work on the old revision's workflow stops.

RA bill and JMR (Stretch) A running account (RA) bill is a contractor's part-payment claim for one milestone. A joint measurement record (JMR) is the quantity of work measured on site and signed by both sides, one of the proofs the milestone needs.

ONE RECORD, END TO END

Drawing AIL-DHJ-STR-DWG-0103 Rev A: an overdue step, a chase and a delegate

- 1 Inbox email 0268 (30 Sep 2026, 09:45) carries transmittal JIL-AIL-TR-0338: Jyoti Infraprojects sends AIL-DHJ-STR-DWG-0103 Rev A under work order AIL/DHJ/WO/B-011, and Document_Register lists it as the drawing's last_transmittal.
- 2 Approval_Chains chain AC-DHJ-STR-4 (B-011, Structural) sets the order: step 1 Ronak Chauhan (Design Lead, 5 days), step 2 Aarav Verma (Package Manager, 4 days), step 3 Pooja D'Souza (Client Representative (PMC), 7 days).
- 3 Workflow_Steps WF-004481 shows step 1 approved by Ronak Chauhan on 2 Oct and step 2 sent to Aarav Verma on 2 Oct at 16:45, due 06-10-2026, still Pending.
- 4 At 9 Oct 2026, 18:00 the step has been overdue since 7 Oct, and Aarav Verma has no entry in Approver_Availability, so the AI Employee chases him; the system reminder in inbox email 0343 drew no reply.
- 5 The chase releases staged reply 0001 (see mailbox_index.csv): Aarav Verma approves Rev A by email with no comments and asks document control to record it from the mail.
- 6 The AI Employee records step 2 as Approved, then checks step 3: Approver_Availability shows Pooja D'Souza on leave from 30 Sep to 15 Oct with Aakash Kothari as her recorded delegate.
- 7 Step 3 is sent to Aakash Kothari, and the drawing stays in approval at step 3.

Outcome. Chase Aarav Verma for step 2 (Package Manager, due 6 Oct). Once his approval arrives by email, record step 2 as Approved and send step 3 (Client Representative (PMC)) to Aakash Kothari, the recorded delegate for Pooja D'Souza.

Why. Clause 3.2 makes the step overdue from the day after its due date, clause 3.1 lets step 3 go out only after step 2 is actioned, and clause 4.1 lets a delegate recorded in the calendar act during the leave. The approvals are human decisions that the AI Employee only records; no escalation to the Project Director is needed because both people at the open steps are available or covered.

WHAT THE ROUND 1 MINIMUM NEEDS

Tables	drawing_register (Document_Register), workflow_step (Workflow_Steps), approval_chain_step (Approval_Chains), approver_leave (Approver_Availability). Everything else in the pack is optional for the minimum, and every table is ready to import from setup_kit/ in this folder.
Staged replies	Staged reply 0001 (the approval in the record above), and the Project Director's reply naming a delegate if you declare the approver on leave.
Watch for	Send your own chase: the workflow system's reminders in the mailbox do not count. Project invoices are RA bills, and they are Stretch.

WHAT THE JOB COSTS BY HAND

196 transmittals arrived between 04 Jul and 09 Oct 2026, about 60 a month. Time one by hand: minutes per record times records per month is what your AI Employee gives back.